

Python Course for Physicists (And everyone else!)

Chunk 5: Loops

By Jonathan Bollig

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Loops: The Idea

- Problem without loops: Print every element in a list individually.
- Easier way: for – loops: Iterate through “iteratable” object: Lists and Set (later more)

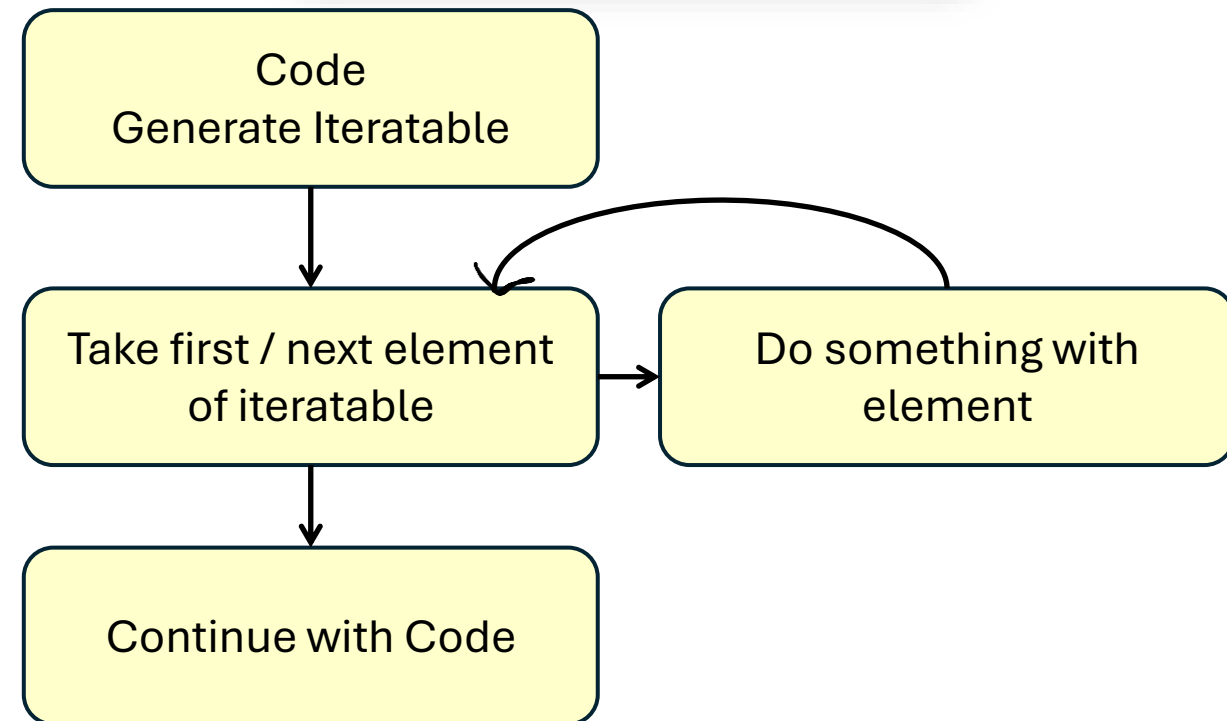
- Syntax:

```
my_list = [100, 50, 0, 99]
for element in my_list:
    print(element)

100
50
0
99
```

```
my_list = [100, 50, 0, 99]
print(my_list[0])
print(my_list[1])
print(my_list[2])
print(my_list[3])
```

```
100
50
0
99
```



Live Coding: Loop examples

- Squaring numbers:

```
numbers = [1, 2, 3, 4, 5]
for i in numbers:
    print("The square of " + str(i) + " is " + str(i**2))
```

```
The square of 1 is 1
The square of 2 is 4
The square of 3 is 9
The square of 4 is 16
The square of 5 is 25
```

- Filtering strings:

```
mixed_data = [1, 4, 3.0, "Hello", 2, "three", None, True]
print("Strings:")
for cake in mixed_data:
    if type(cake) == str:
        print(cake)

print()
print("Other type of data:")
for cake in mixed_data:
    if type(cake) != str:
        print(cake)
```

Strings:

Hello
three

Other type of data:

1
4
3.0
2
None
True

Live coding: range and zip

- Range: Iterable that counts:
`range(n) ≐ [0, 1, 2, 3, ..., n-1]`

```
for i in range(5):  
    print(i)
```

0
1
2
3
4

- zip: iterate multiple iterators:

`for i, j in zip(list1, list2):`



```
names = ["Dieter", "Felix", "Mareike", "Lea"]  
ages = [43, 4, 23, 29]  
for name, age, in zip(names, ages):  
    print(name + " is " + str(age) + " years old".)
```

Dieter is 43 years old
Felix is 4 years old
Mareike is 23 years old
Lea is 29 years old

Caveat: C / Java Style Loops

- Don't use C / Java style loops !!!!!

```
my_list = [100, 50, 0, 99]
for i in range(len(my_list)):
    print(my_list[i])
```

```
100
50
0
99
```



```
my_list = [100, 50, 0, 99]
for element in my_list:
    print(element)
```

```
100
50
0
99
```

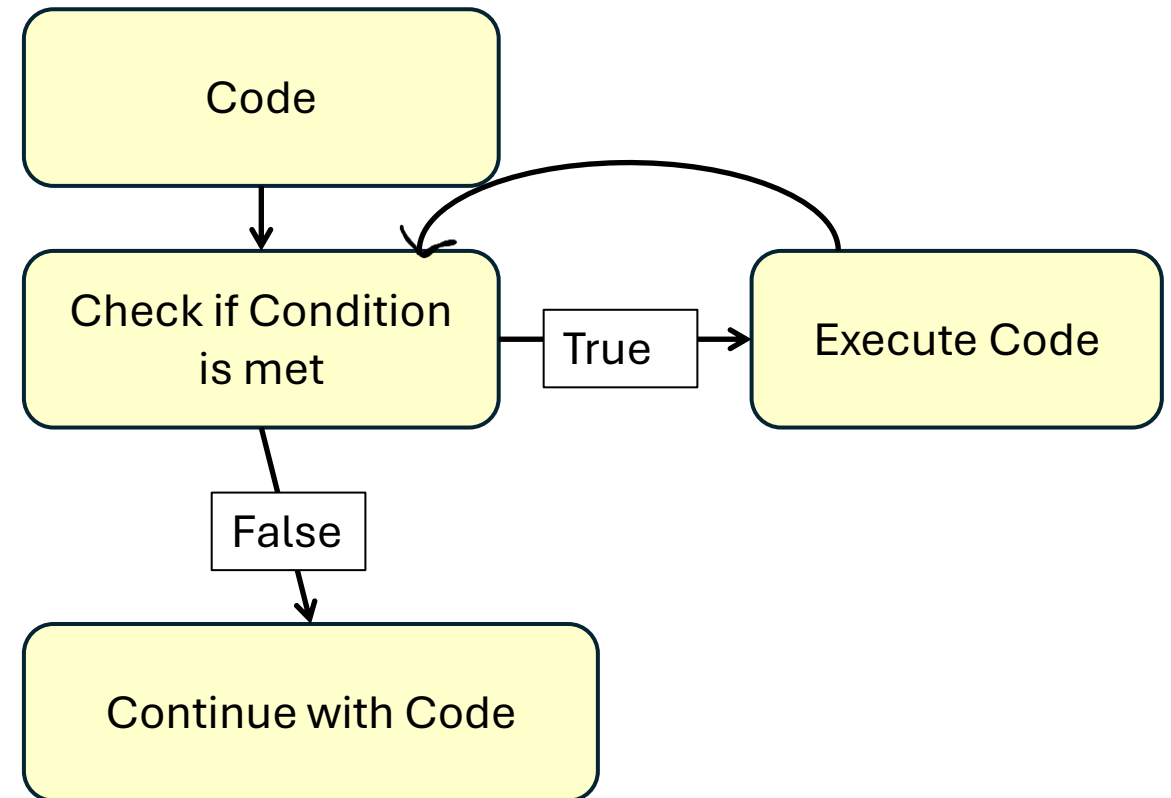


while - loops

- Used, when the number of iterations is unknown → not so often in scientific programming.

```
number = input("Enter a positive number > 20: ")
print(number)
while int(number) <= 20:
    number = input("Try again: ")
print("Success")
```

```
Enter a positive number > 20: 10
10
Try again: 5
Try again: 10
Try again: 5
Try again: 22
Success
```



break & continue

- Rarely used in scientific computing.
- `continue` $\triangleq$ skip current iteration:
- `break` $\triangleq$ exit loop:

```
for i in range(10):  
    if i%2 == 0:  
        continue  
    print(i)
```

1
3
5
7
9

```
data = [1, 4, 2, 9, 4, 5, 8]  
for i in data:  
    if i>6:  
        break  
    print(i)
```

1
4
2

Good Luck with your Tasks 😊



[<https://medium.com/codex/10-best-programming-memes-1-cec344afc19a>]